

Measurement-shifting in the syntax: A rational solution to an agreement-interpretation puzzle?

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24 July 2026

1 Introduction

This work centers around a puzzle concerning FRACTION PARTITIVES (FPs)—a nominal construction exemplified in (1)—and their syntax-semantics. FPs which embed a plural DP are canonically understood to refer to a subset of their complement whose cardinality is that of the complement multiplied by the fraction. Ergo, (1) could mean two out of six apples, three out of nine apples, etc. (see, e.g., [Ionin et al. 2006](#); [Wagiel 2021](#); [Benbaji-Elhadad and Wehbe 2024](#)).

(1) One third of the apples

For languages like American English, this understanding is fairly suitable. However, the FPs of some languages exhibit an additional reading which has been discussed in [Hoeksema \(1996\)](#); [de Hoop \(1997, 2003\)](#); [Falco and Zamparelli \(2019\)](#). To illustrate, imagine a dingy studio apartment which aerially resembles an unequilateral hexagon, with two four-meter-wide walls and four one-meter-wide walls. Assuming the walls are of uniform height, the combined surface area of the apartment’s walls is $12 \times h$ square meters (where h is the apartment’s height). This is diagrammed in *fig. 1*.

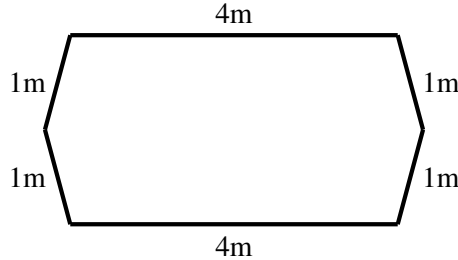


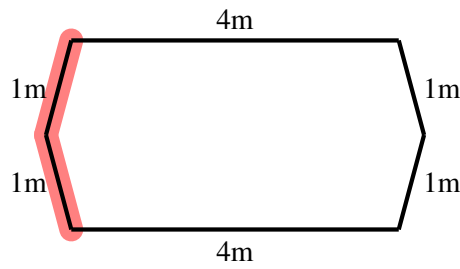
Figure 1: Floor plan of the studio apartment (not to scale)

If an Italian-speaking tenant of such a unit were to report (2), this would be compatible with two scenarios. First, it may be that two out of the six walls are completely covered in mold (*fig. 2a*); this interpretation, which I presented above, is what I call the COUNT reading. Alternatively, it could be that one of the large walls (whose surface area is a third of that of the combined walls) is completely covered in mold (*fig. 2b*); I dub this interpretation the MEASURE reading.¹ Because the latter interpretation is unavailable in American English, the English analogue of (2) is only compatible with the scenario in *fig. 2a*.

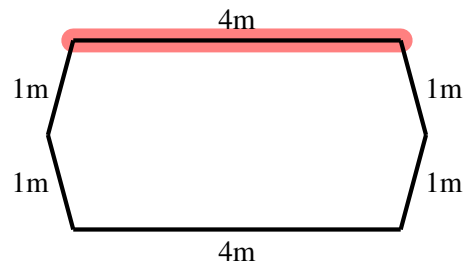
(2) la muffa ha coperto un terz-o del-le paret-i
DET.F.SG mold has.3SG covered.3SG one third-M.SG of-DET.F.PL wall-F.PL
'The mold has covered a third of the walls'

[Italian]

¹My use of *unambiguously* in *fig. 2a, 2b* should be clear; if the apartment layout was such that every member of the wall plurality was congruent (same size and shape), then there would be substantial overlap between the count and measure reading. For example, in a collection of six walls which are each $2hm^2$, two walls (cumulatively $4hm^2$) may be selected by either the count or measure reading. A collection of walls such as that in *fig. 1* guarantees that neither the count nor measure readings will hold simultaneously.



(a) Unambiguously COUNT context



(b) Unambiguously MEASURE context

Figure 2: Two possible scenarios for mold coverage

The puzzle is further enriched by a complicating detail that, as far as I am aware, has gone unnoticed in previous literature. If a FP is in a position of agreement, agreeing elements can copy the features of either the denominator of the fraction (in this case, *terzo* (masculine singular)) or the complement of the FP (*le pareti* (feminine plural)). If the denominator is agreed with, only the measure reading is available. If instead agreement is with the complement, only the count reading is available.

(3) ‘One third of the walls are covered by mold’ [Italian]

- a. (esattamente) un **terz-o** del-le paret-i è coperto di muffa
(exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3SG covered.M.SG by mold
✓M, ×C
- b. (esattamente) un **terz-o** del-le **paret-i** sono coperte di muffa
(exactly) one third-M.SG of-DET.F.PL wall-F.PL be.3PL covered.F.PL by mold
×M, ✓C

Of course, the measure interpretation is compatible with other scenarios as well; the only constraint is that the cumulative square meterage of the moldy region be $4h\text{ m}^2$. It could be that the lower third of each wall is covered in mold; In an even more dire situation where there are only two walls, it could be that two thirds of one wall are covered in mold. I use the dichotomy in *fig. 2a, 2b* for expository purposes.

This general pattern extends to other kinds of count nouns besides just *walls*. The same can be observed with more-transparently discrete pluralities like cakes, and even pluralities of abstract objects like films.

(4) ‘A third of the cakes were eaten’ [Italian]

- a. **un terz-o** delle tort-e è stat-o mangiat-o
a third-M.SG of.the.F.PL cake-F.PL be.3SG been-M.SG eaten-M.SG
✓M, ×C
- b. un **terz-o** **delle** **tort-e** sono stat-e mangiat-e
a third-M.SG of.the.F.PL cake-F.PL be.3PL been-F.PL eaten-F.PL
×M, ✓C

(5) ‘A third of the films were watched by John’ [Italian]

- a. **un terzo** de-i film è stato visto
one third of-M.PL film be.3SG been.M.SG watched.M.SG
✓M, ×C
- b. un **terzo** de-i **film** sono stati visti
one third of-M.PL film be.3SG M.PL watched.M.PL
×M, ✓C

Furthermore, this is not a phenomenon unique to Italian; it can be observed to some extent in many Indo-European languages. French and Greek exhibit exactly the Italian pattern.

(6) ‘One third of the walls are covered by mold’ [French]

- a. **un tiers** des murs est couvert de moisissures
one third of.DET.M.PL wall.M.PL COP.3SG covered.M.SG by mold

✓M, ×C

- b. un tiers **des** murs sont couverts de moisissures
 one third of.DET.M.PL wall.M.PL COP.3PL covered.M.PL by mold
 ×M, ✓C

(7) ‘One third of the pies were eaten’ [Greek]

- a. **ena trit-o** ton pit-on fagoθik-e
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3SG.PST
 ✓M, ×C
- b. ena trit-o **ton** **pit-on** fagoθik-an
 one third-N.SG DET.GEN.PL pie-GEN.PL eat.PASS.PRF.PST-3PL.PST
 ×M, ✓C

A similar pattern can be observed with the fraction partitives of Russian and German, albeit with the caveat that a count reading is still available when Agreement is with the denominator.

(8) ‘One third of these² cakes were eaten’ [Russian]

- a. **(odna) tret’** étix pirog-ov byl-a s”eden-a
 one third.F.SG this.GEN.PL cake-GEN.PL be-F.SG eaten-F.SG
 ✓M, ✓C
- b. (odna) tret’ **étix** **pirog-ov** byl-i s”eden-i
 one third.F.SG this.GEN.PL cake-GEN.PL be-PL eaten-PL
 ×M, ✓C

(9) ‘One third of the cakes were eaten’ [German]

- a. **Ein Drittel** der Kuchen wurde aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3SG eaten
 ✓M, ✓C
- b. Ein Drittel **der** **Kuchen** wurden aufgegessen
 one third DET.GEN.PL cake.GEN.PL be.PST.3PL eaten
 ×M, ✓C

Compare all of these cases again with American English—a language where only the count reading is available. Interestingly, when the complement of a fraction partitive is plural, agreement must necessarily be with that complement.

- (10) One third of the walls {*/is/are} moldy
 ×M, ✓C

The generalization can thus be described as the following. When a DP-external element (e.g., a verb) agrees with the complement of a FP, the interpretation is necessarily a count one. When said element agrees with the denominator (in languages where circumstances allow agreement with a denominator), there is at least a measure interpretation. Whether or not there is also a count interpretation under fraction agreement is a point of cross-linguistic variation.

The primary objective of the present study is to provide an account of this alternation specifically for Italian. Secondly, I seek to explain why the alternation exists in some languages but not others and develop an account as such. In developing an explanation, I ultimately make two claims. Firstly, I argue that the agreement facts under the measure interpretation are the work of a covert operator that blocks agreement with the complement of the FP. Therefore, what we have been calling “fraction agreement” is not Agreement with the features of the denominator; rather, there is an Agree relation between the relevant probe and said operator. My second claim pertains to what this operator does on the interpretive end. Specifically, it is suggested that it is an intensional operator which can affect how variable measure functions are determined. Implementing this solution requires an overhaul of how measurement is defined at LF, a stark departure from previous work on the semantics of measure (which often assumes that the process is deterministic for pluralities and otherwise purely contextual).

²Consultant found the measure reading easier to attain with a demonstrative heading the complement. The measure reading is still marginally available without a demonstrative.

The paper is structured as follows. I establish background knowledge of the semantics of parthood and measurement in service of demonstrating how previous authors have derived the count reading in §2. §3 considers a previous proposal to additionally derive the measure reading. The first claim of this work comes in §4, where I extend the incumbent proposal to account for the agreement alternation. I reexamine said proposal in §5 and ultimately find it untenable for the current suite of Italian data. §6 houses the second claim, where I develop a new analysis to account for the troublesome data. §7 concludes.

2 Deriving Count FPs

Let us take a moment to review how the more-conventional count reading is derived in a formal semantic system per the literature. Descriptively, there are two components to an FP’s semantics. First is a condition of parthood; a FP must pick out a proper part of the individual denoted by the complement.³ Secondly, there is a comparison between a degree of the referent and a degree of the complement on the same scale. The measure of the individual on some scale is the measure of the complement on the same scale divided by the denominator. Intuitively, if a rock is three kilos, a third of that rock will be one kilo. Momentarily putting aside the issue of the measure reading, the main challenge in providing an account of the conventional behavior of FPs is their selectional flexibility. Consequently, the truth conditions of an FP are constrained non-trivially depending on the argument it takes.

Let us first examine the parthood component. Importantly, FPs can take either singular or plural individuals as their complement. This comes as no surprise as it can likewise be observed in many partitive constructions in many of the world’s languages (Moltmann, 1997; Wagiel, 2021). This can be seen transparently for Italian in (11), but it also extends to languages like Passamaquoddy (Algonquian), which is very much unlike the European languages we’ve been considering up to this point.

- (11) un terzo {della mel-a / delle mel-e}
 a third {of.the.F.SG apple-F.SG / of.the.F.PL apple-F.PL}
 ‘A third of the apple/apples’

- (12) a. epahsiw cikon wekine
 half apple.PROX.SG spoil.AI
 ‘half of the apple spoiled’
 b. epahsiw cikoni-yik wekini-yik
 half apple-PROX.PL spoil.AI-PROX.PL
 ‘half of the apples spoiled’

[Passamaquoddy]

Thus, a formal definition for FPs requires a shared type between singular and plural DPs. I follow Link (1983) et seq. in treating pluralities as individuals (type *e*) rather than as sets ($\langle e, t \rangle$, cf. Bennett 1974; van der Does 1991; Winter 2001). Borrowing heavily from Classical Extensional Mereology (CEM, Lesniewski, 1931; Leonard and Goodman, 1940), he treats pluralities as mereological sums of “atomic” individuals (singularities), which are of course also type *e*.

- (13) a. $[[\text{wall}]]^c = \{a, b, c, d\}$
 b. $[[\text{the walls}]]^c = \bigoplus [[\text{wall}]]^c = a \oplus b \oplus c \oplus d$

This nets the selectional flexibility for FPs that we seek; they are compatible with both singularities and pluralities by virtue of their being the same semantic type (*e*). However, adopting Link’s proposal wholesale for an account of FPs leaves us with a problem. In this system, membership to a plurality is modeled with the mereological relation (PROPER) PART-OF ($\sqsubset$).⁴ However, Link distinguishes two kinds of parthood which relate different kinds of objects. One, INDIVIDUAL *part-of*, holds between pluralities and the (sums of) individuals which compose them. The other, MATERIAL *part-of*, holds between singular individuals and their parts. This is done in service of the observation that predicates can apply to singularities but not necessarily the sum of their parts; as the famous example goes, a pure gold ring may be new even if the gold which composes it is old. So, while both

³Some works (e.g., Ionin et al. 2006; Magri 2008) present an analysis of partitives s.t. *of* and its equivalents are the locus of the parthood assertion. While this is a plausible analysis, I do not follow this in the present work because—as far as I am able to tell—it confers no analytical advantage and defining fractions with parthood baked-in is notationally simpler.

⁴In a system that treats pluralities as sets, this would naturally be captured with a set-membership or (proper) subset relation.

pluralities and singularities are type e , following Link’s system would predict that FPs—which include parthood in their meaning—could only be compatible with one type of argument or the other, else fractions are ambiguous between a singularity-selecting form and a plurality-selecting form.

This is difficult to reconcile with Moltmann and Wagiel’s observation about partitive constructions, prompting them to abandon the distinction between individual and material part-of.^{5,6} For the same reasons, I follow suit. Now, $\sqsubset$ is what holds between my material parts and me as well as me and any plurality of which I am a member. In lieu of *atoms*—which Link defines as individuals without proper individual parts, an empty designation in a system which collapses material and individual part-of—Moltmann and Wagiel define the basic individual which constitutes a singularity as an INTEGRATED WHOLE (IW).⁷ Pluralities are therefore sums of IWs.⁸

With parthood out of the way, let us consider how to model the comparative component of fractions. To begin, I introduce to our ontology DEGREES (type d), which we may conceptualize as points on scales of measurement. Many natural language phrases arguably denote degrees, like heights (2 *meters*), monetary values (\$38), temperatures (21°C), and even cardinalities (*twelve*). It is standardly assumed that degrees can participate in arithmetical relations like *greater than*, *less than*, *equal to*, etc, and that at least some arithmetical operations (addition, subtraction, multiplication) may apply to degrees as well. I further assume, following Coppock (2022), that degrees may participate in division (cf. Bale and Schwarz 2022, 2026). Degrees are retrieved from individuals via MEASURE FUNCTIONS (MFs, $\langle e, d \rangle$). There are a number of natural language items which plausibly involve measure functions, most prominently gradable adjectives and measure words. Sample denotations for such items are given in (14), (15).

$$(14) \quad [[\text{tall}]] = \lambda d_d \lambda x_e. \mu_{\text{height}}(x) \geq d$$

$$(15) \quad [[\text{weigh}]] = \lambda d_e \lambda x_e. \mu_{\text{weight}}(x) \geq d$$

In these cases, the scale on which the argument of the measure function is evaluated is lexically-specified; *weigh* returns a degree on the weight scale, for example. However, a semantic analysis of fraction partitives demands something less specified. Forgetting momentarily about partitives with plural complements, the measure function of a fraction must be somewhat flexible; compare *a third of the apple*, where the relevant dimension of comparison could be volume or mass, and *a third of the movie*, where the dimension is likely runtime. Even the same argument can be measured in distinct ways depending on the situation; compare the two uses of *half of the apple* in (16).

⁵For additional work which assumes a single parthood relation for individual and material parts, see Magri (2008); Sudo (2025).

⁶Moltmann’s view on what a part can be goes beyond what I discuss in this section. She also notes that in certain cases it can depend on the whole itself what the parts are, which is explored in Moltmann (1998, prep)

⁷How an integrated whole is defined varies between the two authors. Moltmann appeals to R-INTEGRATED WHOLEs (Simons, 1987), which are defined in (i). To Wagiel, an IW is an individual that has the mereotopological property of being MAXIMALLY-STRONG SELF-CONNECTED (Casati and Varzi, 1999) relative to some property. Essentially, an MSSC individual is one who is SELF-CONNECTED (any two ways of dividing an individual are connected), has an interior which is self-connected, and the self-connected things which overlap with it are part of it. A formal definition is given in (iv).

- (i) An entity x is an R-integrated whole if there is a division of x such that every member of that division stands in the relation R to every other member and no member bears R to anything other than members of the division. (Moltmann, 1997, 24)
- (ii) Self-connected

$$SC(x) \stackrel{\text{def}}{=} \forall y \forall z [\forall w (\text{overlap}(w, x) \leftrightarrow (\text{overlap}(w, y) \wedge \text{overlap}(w, z))) \rightarrow \text{connected}(y, z)]$$
(x is self-connected iff any two parts that form the whole of x are connected to each other.)
- (iii) Strongly self-connected

$$SSC(x) \stackrel{\text{def}}{=} SC(x) \wedge SC(\text{interior}(x))$$
(x is strongly self-connected if x is self-connected and the interior of x is self-connected.)
- (iv) Maximally-strongly self-connected

$$MSSC(x) \stackrel{\text{def}}{=} SSC(x) \wedge \forall y [SSC(y) \wedge \text{overlap}(y, x) \rightarrow y \sqsubset x]$$

⁸Since parthood is generally assumed to be a transitive relation (Varzi, 2003), this system produces seemingly problematic predictions about what may be part of a plurality. I eschew an extended conversation of this topic for sake of relevance, but Moltmann and Wagiel patch this problem by additionally positing that transitivity may not hold across individuals which are integrated wholes (i).

(i) $\forall x, z. x \sqsubset z \rightarrow \neg \exists y. x \sqsubset y \wedge y \sqsubset z \wedge \text{int-wh}(y)$

- (16) a. Paint half of the apple $\rightsquigarrow$ surface area
 b. Use half of the apple for the pie $\rightsquigarrow$ volume, mass

In this sense, fraction partitives resemble *more*-comparatives, which have been subject to considerable inquiry. As can be seen in (17), *more* is used in a comparison of degrees from two different scales. To capture this fact, it and other quantity words (like *much*) have been analyzed as introducing a VARIABLE MEASURE FUNCTION (VMF, represented μ^9) whose scale of measurement is not specified when it enters the derivation (Bale and Barner, 2009; Wellwood, 2019). Wagiel ultimately adopts μ for his account of the semantics of FPs.

- (17) a. Newman spun more yarn than Jerry $\rightsquigarrow$ length
 b. Kramer listened to more music than George $\rightsquigarrow$ runtime

How VMFs are defined is not a free-for-all, though. Two important constraints have been identified to produce a successful analysis of, e.g., quantity words. For starters, the resulting MF must be MONOTONIC as defined in (18) (Schwarzschild, 2002, 2006; Nakanishi, 2004, 2007; Wellwood, 2019). In prose, a monotonic MF is sensitive to mereological relationships—a proper part of an individual measures less on a monotonic scale than the whole. By this definition, volume, weight, length, runtime, and cardinality are all monotonic scales. On the other hand, temperature, density, concentration (of, e.g., alcohol or salt in a solution), and speed are non-monotonic scales. Such scales ought to be excluded for defining VMFs given the kinds of interpretations comparative constructions like (19) can have.¹⁰

- (18) S-monotonicity Adapted from Wellwood (2019)
 A measure function μ_m is S-monotonic if $\forall v, v'. v \sqsubseteq v' \rightarrow \mu_m(v) \leq \mu_m(v')$

- (19) Jerry drank more wine than Elaine
 a. ✓ Jerry drank two glasses while Elaine drank one
 b. #Jerry's wine was 45°F while Elaine's was 42°F

Let us now turn to the second constraint. Bale, Barner, and Wellwood propose that if the argument of a VMF is a plurality, the resulting MF must be a cardinality function. This position is motivated by a series of observations about how pluralities are interpreted in comparative constructions across a variety of languages and morpho-syntactic contexts. Experimental studies by Barner and Snedeker (2005) and more recently Le et al. (2025) have found that collections of discrete objects are consistently evaluated by number (rather than size or weight) in quantity comparison tasks. This is true even of object-mass nouns like *silverware* or *furniture*, which are like more prototypical mass nouns in refusing plural morphology and numerals. These results extend to languages like Japanese and Mandarin, where (morphological) mass-count and singular-plural distinctions are not as clear-cut as they are in English (Inagaki and Barner, 2009; Cheung et al., 2010).;

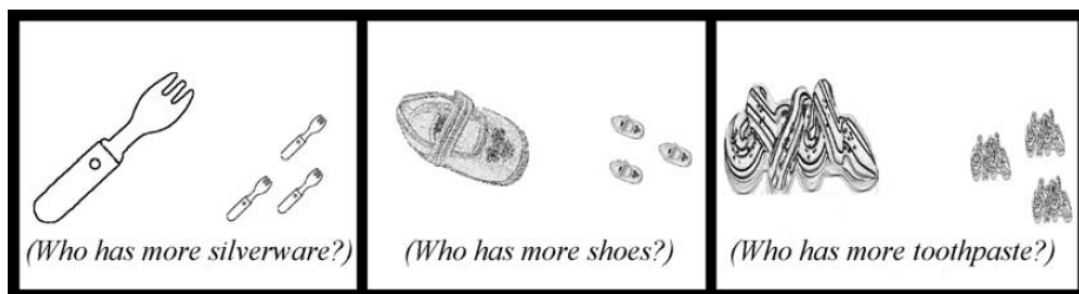


Figure 3: Experimental stimuli from Barner and Snedeker 2005, p. 50

This can be observed for pluralities of events as well as entities. (Wellwood, 2019, ch. 6) notes that certain

⁹...without a subscript.

¹⁰This constraint extends beyond defining VMFs. As Krifka (1989) notes, heads of measure pseudopartitives can only be monotonic (also called EXTENSIVE) measure functions.

- (i) {five ounces/*twenty carats} of gold

“high-attachment” comparative constructions evoke a reading that is sensitive to the number of occasions rather than degree on some other scale. This data might encourage us to bias VMFs towards cardinality functions over other forms of measurement.

- (20) Ann was happy more than Bill was.
 ✓ more occasions, #more happiness

Bale & Barner implement these constraints as follows. They posit a partially-ordered set of measure functions which is composed of every monotonic measure function. Furthermore, cardinality is outranked by every other form of measurement in this set. This set, which I call M , is shown in (21).

$$(21) \quad M = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

Bale & Barner further posit an algorithm for determining the scale of a VMF from the set of functions M . Defined in (22), a VMF is defined as the minimal MF in M which ranges over the argument it has taken. If there is no single minimal MF, context decides which MF it will be defined as. Since a cardinality function necessarily ranges over pluralities by making reference to atoms/integrated wholes, this algorithm guarantees a plurality will be counted when it is fed to a VMF.

- (22) Contextual conditioning
 μ is interpreted as one of the measure functions m_z in the series $\langle m_1, m_2, m_3, \dots, m_n \rangle$ such that the argument for μ is in the range of m_z ; furthermore, contextually m_z is preferred to m_y if $m_z < m_y$

Let me now briefly demonstrate how this system works. Imagine a VMF has taken an atomic individual/integrated whole like an apple as its argument, as in (23). This individual has many properties we expect a physical object to have—it has a weight, height, width, surface area, density, volume, and so forth. Any of the measure functions which correspond to these concepts could be applied to the individual apple and return a defined, nonzero degree. However, there are measure functions which do not range over the individual. It obviously does not have a runtime, for example. For our purposes, we will further stipulate that a cardinality function cannot range over it.¹¹ We now take the set of MFs which are defined for the apple and intersect it with M (21), the result shown in (24).

$$(23) \quad \mu([[\text{the apple}]]^c)$$

$$(24) \quad \{m : m \in M \wedge m([[\text{the apple}]]^c) \neq \# \} = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

The resulting set contains multiple MFs, none of which are minimal with respect to one another. Thus, context will ultimately pick the identity of the VMF from the set of remaining candidates. Once it has been defined, the function will return a degree along the relevant scale.

How do things play out when the argument of the VMF is a plurality? Consider such a case with a plurality of apples, for example. Once again, we intersect M with the set of MFs which are defined for the plurality. Importantly, the plurality is still defined for non-cardinal MFs; this is irrefutable given how lexically-specified measure functions can combine with pluralities and have cumulative interpretations (25). Yet, they also have a cardinality by virtue of being a plurality. The resulting set (26) now has a single minimal member, the cardinality function. Per the algorithm, the VMF will necessarily be a cardinality function. Context no longer has a say in the matter.

- (25) The chairs (altogether) weigh 200 kilograms.

$$(26) \quad \{m : m \in M \wedge m([[\text{the apples}]]^c) \neq \# \} = \{\mu_{card} < \mu_{weight}, \mu_{volume}, \mu_{mass}, \mu_{length}, \mu_{runtime}, \dots\}$$

¹¹It is too strong to say that a singularity like the apple does not have a cardinality. If Kramer only has a single apple, an inquiry into the degree of apples Kramer owns will obviously return the value *one*. In the original formulation of this algorithm, VMFs combine with sets $\langle (e, t) \rangle$ rather than individuals (e); cardinality functions in particular are only defined for join semi-lattices which include atomic individuals (Bale and Barner, 2009, §5.3.1). This rightfully excludes bare NPs (which are sets of atomic individuals but not their sums) and mass nouns (which are atomless semi-lattices, being divisible *ad infinitum*.) Wagiel (2021)’s adaptation treats MFs as taking a type- e argument, which is necessary if we believe fractions are combining with DPs but obfuscates the straightforward singularity-plurality distinction. We can patch this by additionally stipulating that the identity of a VMF must range over the arguments of every MF in the FP; cardinality is defined for a single apple but not a part of it, whereas cardinality is defined for a plurality of apples as well as the IWs and arbitrary sums of IWs which compose it.

With this machinery in place, I define fractions like *third* in (27). It takes an individual y and returns a set of individual x s with two constraints imposed. Firstly, x must be a proper part of y . Secondly, the degree of x must equal the degree of y multiplied by the fraction (in this case, $\frac{1}{3}$). The identity of the relevant degree is subject to the algorithm I sketched out above. To combine this set with predicates, it may first be taken by a choice function or ι -operator (producing type e) or combining with an existential quantifier ($\langle\langle e, t \rangle, t \rangle$). The latter of these is most likely what is observed in more prototypical uses of fractions, given what appears to be an indefinite article (*a, un, ...*) at the left-edge.¹²

$$(27) \quad [[\text{third}]] = \lambda y_e \lambda x_{e.x} \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3}$$

How might we derive fractions with numerators greater than one (e.g., *two thirds*)? One option would be to treat the fraction as any other countable NP. First, we apply Link (1983)’s $*$ -operator to the set, yielding a join semi-lattice of lone thirds, arbitrary sums of those thirds, and the maximal sum of the thirds. The set could then combine with a numeral (*two*), restricting the set to only entities which are the sum of exactly two “third”-entities. The second option would be to abstract over the numerator and have the fraction take a numeral as an additional argument. The latter of these options is attractive since there seem to be at least some uses of fractions where the numerator and denominator form a constituent (28). However, this does not provide a straightforward explanation of the plural morphology seen on fractions with numerators ≥ 2 , unlike the former approach. I do not commit to one analysis over the other at this juncture, but I highlight both as possible options.

(28) The company hired [two-thirds] New Yorkers

To sum up this section, we laid out the background assumptions motivated by findings from previous literature to provide a semantics for the more-conventional count reading of FPs. Happily, this analysis is flexible enough to deal with both singularities and pluralities. The parthood component can be handled with tools from mereology and the assumption that pluralities and singularities may be of the same semantic type. Contextually-defined measure functions which are additionally biased towards counting for pluralities characterizes the degree comparison component. Specifically, the count reading follows because of the ordering of the set in (21). Of course, these are insufficient for capturing the measure reading we observe in Italian, etc. The architecture for defining variable measure functions is reconsidered when I present my own proposal in §5. In the following section, though, we examine a prior suggestion for deriving the measure interpretation.

3 Measure FPs: Covert Count-Mass Coercion?

Let us now turn our attention to the measure reading and how the “incumbent” proposal derives it. Beforehand, though, I present a chronology of formal thought on the measure reading. While the fact that the measure interpretation is tracked by agreement in some languages is a novel contribution of this work, the reading itself has not gone unnoticed. Something like it has been suggested at least since Hoeksema (1996) who notes that a FP like *half of Jane and Jacky* most naturally refers to the sum of half of Jane and half of Jacky rather than one of them individually.¹³ However, the first attempt at an account was not until de Hoop (1997, 2003). In her account of the Partitive Constraint, she proposes that (most) plural DPs are ambiguous between individuals (e) and sets ($\langle e, t \rangle$). She further posits that some partitives differ in what kind of semantic type they select for; most relevant to us, the heads of proportional partitives (fractions, but also percentages, etc.) can take either type as an argument. Thus, the difference in meaning is resolvable to the semantic type of the complement.

¹²At least in English, introducing a universal quantifier to a sentence as in (i) also containing an FP permits both co-varying and non-co-varying readings. This lends additional credence to the quantifier option.

(i) Every member of the audience punched a third of the comedians.

¹³I must note at this point that these explorations of the interpretation of FPs cite English data. For cases involving conjunction, the preferred reading is very plausibly a case of conjunction reduction (*half of Jane and half of Jacky*) as Falco and Zamparelli (2019) note. Thus, it is distinct from the measure reading we’ve discussed so far. With regards to examples like *half of the flowers*, which F&Z report as having the measure reading, I personally disagree with many of their judgments for my own English. I have observed that the measure reading for FPs is not robustly attested in American English, but it would be interesting to conduct a large scale survey to see if any Englishes do have the measure reading, and along what lines they fall. Such a project is beyond the scope of this paper. Regardless, I discuss the works because their descriptions of the phenomena match that of measure readings and their proposed solutions are plausible approaches to analyzing measure FPs.

However, [Falco and Zamparelli \(2019\)](#) (authors of this section’s topical suggestion), present a number of arguments against this position. Firstly, such an ambiguity for plural DPs makes predictions outside the domain of partitives which are largely unfounded. Treating definite plurals as $\langle e, t \rangle$ opens them up to semantic processes typically reserved for NPs and predicates (quantification, predicate modification, etc.), making forms like **red the flowers* possible without further stipulation. Furthermore, what has the privilege of $e \sim \langle e, t \rangle$ ambiguity in her system is ill-defined. She argues that coordinated singulars can only be e , yet a coordinated DP like *the North Pole and the South Pole* picks out the same plurality as *the Earth’s poles*, a definite plural—which can be either e or $\langle e, t \rangle$ according to de Hoop. For these reasons, I agree with F&Z that this approach is largely untenable.

What kind of an explanation do F&Z offer instead? Their suggestion pairs especially well with the method for deriving count FPs sketched in §2. Recall that the reason the measure reading was ungenerable there was the bias to counting in defining VMFs; μ will be a cardinality function if it takes a plurality as its argument, per the algorithm proposed by Bale & Barner. If we want to hold this constraint constant, one play is to “trick” the fraction into thinking its complement isn’t really a plural individual. Specifically, F&Z suggest that the measure reading is due to a covert operator taking the plural complement and returning unindividuated mass—perhaps the UNIVERSAL GRINDER of [Pelletier \(1975\)](#); [Pelletier and Schubert \(1989\)](#). They write (p. 36):

The only remaining ingredient to obtain mass proportions is the presence of a count-to-mass type-shift (from an object to its material constituent part), which seems to be independently needed in light of many mass uses of count nouns, well documented in the literature[.]

We can thus posit a grinder which takes an individual x (plurality or otherwise) and returns a new individual which is the sum of the material parts of the integrated whole(s) which comprise x . With the individuation structure of the plurality destroyed, it is now impossible to count the members of a plurality (i.e., such an individual cannot have a cardinality). As a result, when this object is the argument of a variable measure function, its identity can not be defined as a cardinality function. When such an individual combines with a fraction as in (29), the set is therefore populated by proportions of the wall-plurality which are a third along some non-cardinal dimension; i.e., we get the measure reading.

(29) $[[\text{terzo}]](\text{UG}(\oplus[[\text{parete}]])$

The appeal of this proposal is that, at first blush, it requires virtually no additional assumptions. It allows us to maintain the algorithm for defining variable measure functions, which seems robust given Bale & Barner and Wellwood’s observations as discussed in §2. Furthermore, the exceptional reading—the measure interpretation—is derived via an off-the-shelf mechanism which may be independently needed for count-to-mass coercion processes. This is all done without positing a type ambiguity for plural DPs as de Hoop does.

I revisit the efficacy of the F&Z’s suggestion for the Italian data in §5. There is now the question of how the novel observation—that this interpretation is tracked by agreement in some languages—may be accounted for under this proposal. The next section introduces the first of this work’s original claims: the agreement characteristic of the measure interpretation is the result of an intervention effect.

4 Proposal I: Measure FP Agreement as Intervention Effect

¹⁴Here, I extend Falco & Zamparelli’s proposal to derive denominator agreement under the measure interpretation. Specifically, I would like to suggest that the observed facts are due to the operator intervening between the

¹⁴There is a superficial similarity between so-called *semantic agreement* phenomena where the verb’s exponents affect how the subject is interpreted exemplified (e.g., (i)) and the alternation (3) exemplifies. A recent and noteworthy account of semantic agreement from [Charnavel and Sportiche \(2025a,b\)](#) posits that the features on agreeing heads (e.g., T/Infl) are at least sometimes semantically-contentful rather than uninterpretable copies of features. Specifically, these features introduce an additional presupposition about the subject and the meaning of the overall sentence is calculated with this contribution.

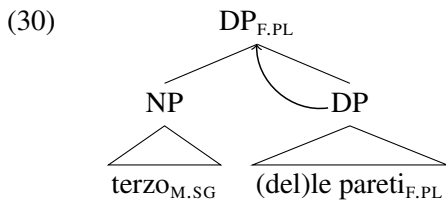
- | | | | |
|-----|----|---|-------------------|
| (i) | a. | The committee is old | [British English] |
| | | $\rightsquigarrow$ the organization was formed a long time ago $\vee$ the members of the committee are old | |
| | b. | The committee are old | |
| | | $\rightsquigarrow$ the members of the committee are old, $\nrightarrow$ the organization was formed a long time ago | |

Charnavel & Sportiche do not extend their account to the count/measure alternation for FPs (nor do they discuss the phenomenon), but one may wonder whether the two deserve a unified account. While their outward similarity is admittedly encouraging, I do not pursue the possibility for the simple fact that both measure and count readings are available for fraction partitives outside of positions of agreement

complement and the probe. This configuration thereby bleeds agreement with the complement and the features of the operator are copied instead. Conversely, syntactic circumstances are such that the complement is agreed with in the unmarked case (the count reading).

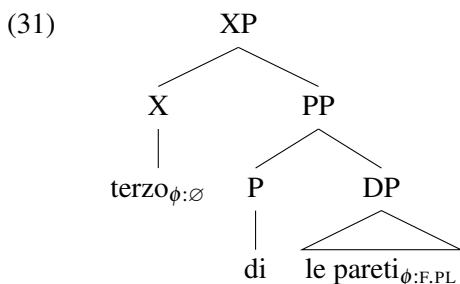
The latter of these points rightfully warrants discussion since, under standard Minimalist assumptions about the Agree operation, competition between multiple bundles of features is resolved by locality (Chomsky, 1995). If the fraction embeds its complement, it is structurally-higher than the complement, and locality therefore predicts that the features of the fraction's denominator should be agreed with. Clearly, this is the opposite of the desired configuration. However, there are two means by which the complement could be agreed with; both ways and their respective issues are discussed in turn.

Firstly, assuming a model where features on a head project to the maximal projection, it could be that the complement is actually heading the FP. This has been suggested for a number of partitive constructions as early as Selkirk (1977), and it gives the correct characterization of the agreement facts without refuting the “featurefulness” of fractions; in Italian and beyond, fractions seem to take number and gender suffixes. Having the complement project respects this fact while getting the correct predictions.



However, there are a number of problems with this approach. Positing such a structure for FPs is unusual given that the complement appears to be introduced to this construction via the preposition *di*. Ergo, to say the complement DP projects its label would necessitate that the preposition projects no syntactic structure. This is not necessarily the death knell as it could be explained as a case-marker in the style of Chomsky (1970), but this begs the question of where this *di* case comes from. If genitive/partitive case is typically assigned by a nominal to its complement, the fraction is the clearest candidate for assigning case. Yet, the present circumstances preclude case assignment by the fraction because the FP complement is the thing which projects (ergo, the fraction does not actually take a syntactic complement).

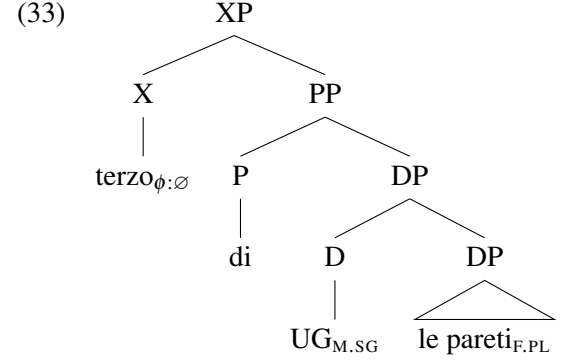
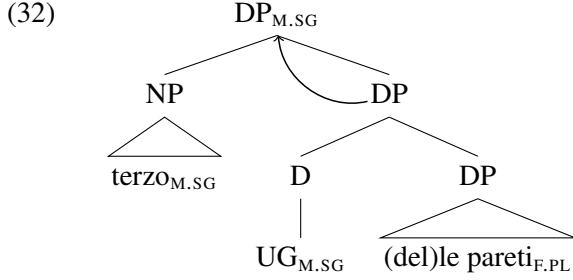
An alternative would be to say that the fraction is blind to agreement processes in some way, in either bearing no features at all or lacking a feature a probe on T can interact with (Deal, 2015). With this configuration, a probe will continue to look down into the FP structure, eventually finding and copying the features on the complement. This derives complement agreement while maintaining a more-familiar structure for partitives. Klockmann (2018) suggests such an analysis for English quantity pseudo-partitives which appear to host number morphology such as {*a lot/lots*} of *New Yorkers*. In these cases, agreement is always with the internal nominal rather than the quantity word.



This nets the desired result but at the expense of a straightforward explanation for morphology on the denominator. Furthermore, we do not presently have good reason to believe that this number morphology is not the number/gender morphology we see on typical nominals. As Klockmann demonstrates, there is strong evidence for *-s* on *lots/tons of NP* not being a genuine plural suffix. For one, the distinction between *a lot of* and *lots of* is semantically arbitrary; we cannot confidently conclude this for fractions since the number suffix is tightly controlled by the numerator of the fraction (**un terzi*, **due terzo*). We also observe the indefinite article *un* showing concord with the fraction (**una terzo*), a fact that is hard to explain if fractions are truly featureless.

(see (2).) Under a Charnavel & Sportiche-type account, it is the features on the agreeing head which licenses additional interpretations. At least one of the two readings would therefore be inaccessible outside of subject position, yet this is not realized.

I remain agnostic about the internal structure of count FPs. What is crucial is that they are configured such that the features of the complement are copied. In order to derive the agreement pattern under the measure interpretation, I posit that the responsible operator is a silent determiner which takes the complement to form a DP. This item enters the derivation in Narrow Syntax with a masculine feature (the same feature which the denominator has or appears to have) that may be copied by a probe on T/Infl. As a result, what resembles agreement with the denominator is actually agreement with the covert operator (which is of course more local to the probe than the base DP). The measure FP structure is schematized for both possibilities in (32)-(33). This way, we may successfully derive the two interpretations and their corresponding agreement patterns when they are in a position of agreement.



Thus concludes my expansion of F&Z’s proposal to fit the Italian empirical picture. To briefly recap, a covert operator akin to the hypothesized Universal Grinder takes a plural individual and returns a new individual which denotes said plurality’s material parts. By virtue of being a mass of unindividuated stuff, the new individual cannot have a cardinality, and any measure function which ranges over it cannot be a cardinality function. Consequently, a fraction partitive which takes such an individual as its argument must have a measure interpretation. On the syntactic side, this operator bleeds complement agreement by virtue of coming with a masculine feature. This proposal is crucially dependent on the count FP being the default case, the syntactic and semantic traits of the measure FP following from the application of the operator to the count structure. So as not to confuse F&Z’s original claims and the novel contributions of this section, I dub this extended proposal F&Z’.

While successful for the suite of data discussed so far, treating the operator as a grinder creates some problems for this account. In the following section, I identify issues which ultimately prompt a reformulation of the operator’s semantics.

5 Grinders Reconsidered

¹⁵As I must reiterate, F&Z’s proposal is quite alluring given how well it dovetails with the assumptions about how the count reading is derived (§2), all with a mechanism which is independently supported. However, there are three issues which must be considered. To begin with, many languages with a grammatical number distinction on nominals allow mass nouns to be “bare”—i.e., without non-singular number morphology or articles. The same is true for count nouns which have been coerced into having a mass interpretation; in French and English, for example, canonically-count nouns appear bare under a mass interpretation (34)-(35), and the same holds for Italian (36). This is straightforwardly explainable if we assume count-to-mass operations apply to NPs (sets of atoms/IWs) instead of DPs (37) (Rothstein 2009; a.o.). Such an operation, I conjecture, would block the application of at least plural morphology.

- (34) Il y a assez de tapis pour nous quatre
EXPL there have.3SG enough of carpet for 1PL four
‘There is enough rug for the four of us’ [French] (Adapted from Bosque 2024, p. 139)

- (35) **The three squirrels** carelessly wandered into traffic and were struck by a motorist.
Now there is **squirrel** all over the road.

¹⁵For the remainder of this work, I specify the consultant(s) for each originally-elicited Italian/Russian datapoint. Where speakers disagree, I mark the sentence with % and specify the individual’s judgment. Speakers are codified as follows: EF: Enrico Flor; GL: Genia Lukin; VO: Vlad Orlov; LP: Lorenzo Pinton.

- (36) c’-è **banan-a** sul paviment-o
 EXPL-be.3SG banana-F.SG on.the.M.SG floor-M.SG
 ‘There’s banana on the floor’

(EF, LP)

- (37) $[[\text{UG}]] = \lambda P_{\langle e,t \rangle} \lambda x_e. \exists y. y \in \pi_i(P) \wedge x \sqsubset y$

Adapted from Kiss et al. (2021)

However, under a measure interpretation of an FP—which F&Z would analyze as the result of a UG applying to the complement—we observe no change on the morphology of the complement in Italian. It is not impossible that the operator we would need to capture the measure reading is distinct from that which is responsible for more-familiar count-mass conversions. This would predict, though, that mass interpretations of plural DPs could be found elsewhere in Italian if context required it. This is because there is nothing stopping $\text{UG}(\text{DP}_{\text{PL}})$ from occurring outside of a fraction partitive. Similarly, the explanation of the agreement facts in F&Z’ predict masculine singular agreement should be possible with feminine plural subjects. At present, I am aware of no such cases.

Furthermore, at least some speakers of Italian permit measure FPs with predicates which necessarily make reference to atoms. If the measure reading is truly resolvable to a covert shifting of the plural complement to a mass, we expect this kind of reference to be impossible; measure FPs should only be compatible with predicates which are insensitive to atomic structure. This would follow from Falco & Zamparelli’s position on how the measure reading is derived—the members of the plurality cannot be counted because the shift from count to mass makes counting impossible. Ergo, they would say VMFs are not defined as cardinality functions. This is of course the case with canonical mass nouns and in prototypical cases of count-to-mass conversion, shown below.

- (38) a. *two water
 b. #The water is parallel/elliptical/stacked.

- (39) Context: The army of robots were smashed to pieces by the anti-AI protesters.
 a. *They swept two robot into a bin.
 b. #Robot is (each) begging for mercy.

By contrast, Italian measure FPs may combine with lexically distributive predicates. One such predicate is *ellittico* ‘elliptical’, seen in (40). In this case, we observe the predicate agreeing with the features of the fraction; this is of course a known characteristic of measure FPs. A verifying case for (40) is shown in fig. 4; there are four ellipses, one square, and one triangle. However, each ellipsis is much smaller than the square or triangle; their cumulative area is only a third of the the cumulative area of all the shapes; therefore, they will be selected for by the measure reading of *un terzo* but not the count reading, since $\frac{4}{6} \neq \frac{1}{3}$. Despite the fact that this is clearly the measure reading, the predicate *ellittico* can still be true of it. Since *ellittico* necessarily holds of IWs, measure FPs must be able to have IWs as part of their interpretation.

- (40) %un terz-o delle figur-e è ellittic-o
 a third-M.SG of.the.F.PL shape-F.PL be.3SG elliptical-M.SG
 ‘A third of the shapes are elliptical’

(EF:✓, LP:???)

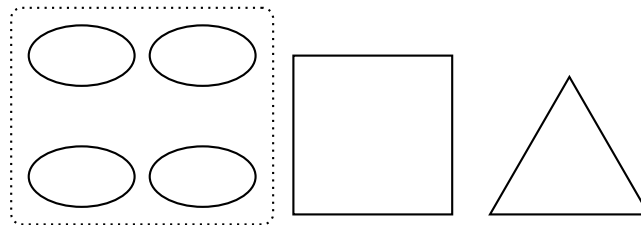


Figure 4: Four ellipses in a collection of six shapes (not to scale).

The same point can be made with predicates containing distance distributivity operators (i.e., derived distributive predicates). In (41), the Italian distributivity marker *ciascuno* ‘each’ forces the predicate to hold of every IW which is part of the subject.¹⁶ As a consequence, that subject must be composed of IWs (i.e., they

¹⁶(41) is complicated by the fact that the distributive operator has feminine agreement. I am not certain why this is, but crucially, the verb agrees with the fraction.

are not masses). Measure FPs (with their respective agreement pattern) may still be subjects of such predicates, further proving that they are indeed pluralities. A verifying case for (41) can be found in *fig. 5*, where four out of six cakes are topped with a strawberry, but said four have a cumulative volume worth one third of the cumulative volume of all the cakes together.

- (41) %un terz-o delle tort-e è decorat-o con una fragola ciascun-a
 a third-M.SG of.the.F.PL cake-F.PL be.3SG decorated-M.SG with a.F strawberry each-F.SG
 ‘A third of the cakes were decorated with a strawberry each’ (EF:✓, LP:??)

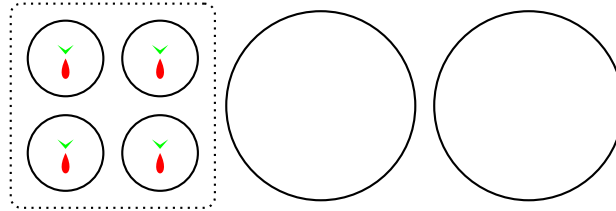


Figure 5: A collection of six cakes, four of which are topped with a strawberry (not to scale).

One final illustrative example comes from discrete collective predicates like *accatastato* ‘stacked’.¹⁷ Such predicates are like those modified by ‘each’ in necessarily selecting for a plurality while making reference to integrated wholes in said plurality; ergo, they cannot be combined with masses. Yet, they may take fraction partitives as their subject while copying the features of the fraction, as shown in (42). This sentence can be true in contexts like *fig. 6*, where four cakes out of six are arranged in a stack, yet these four cakes together have a volume which is one third of the total volume of cakes.

- (42) %un terz-o delle tort-e è accatastat-o
 a third-M.SG of.the.F.PL cake-F.PL be.3SG stacked-M.SG
 ‘A third of the cakes are stacked’ (EF:✓, LP:??)

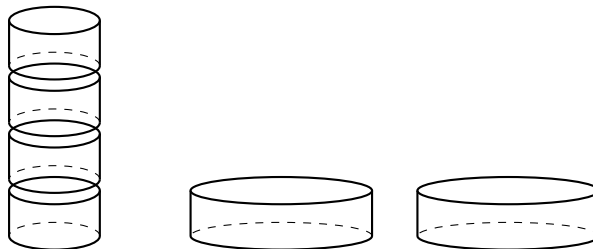


Figure 6: A collection of six cakes, four of which are stacked on top of each other (not to scale).

Were this data not enough, I direct the reader to (43)-(44), two examples taken from the Italian-language corpus PAISÀ (Lyding et al., 2014). In these cases we see FP subjects of predicates which range over (animate) integrated wholes; (44) is particularly interesting because it contains a possessive adjective anaphoric to the individual members of the plurality. In both cases, we see the agreement pattern characteristic of measure FPs. F&Z’s suggestion would predict them to be impossible with these kinds of predicates, but once again this is contrary to fact.¹⁸

¹⁷Astute readers may worry if this particular predicate is genuinely a discrete collective predicate or if it is a mass/plurality-selecting predicate analogous to English *piled (up)*. Observe that this predicate cannot combine with genuine masses like *neve* ‘snow’.

- (i) #la neve è accatastat-a
 the.F.SG snow be.3SG stacked-F.SG
 Int. ‘A third of the snow is piled up’ (EF)

¹⁸With regards to (43)-(44), it is difficult to imagine that the original author meant to pick out a subset of the guards/British women on the basis of anything besides cardinality. For example, if there are 300 guards, the author likely means to say that 100 guards exhibited arguably sadistic behavior. It seems unlikely that a subset of the guards whose cumulative volume or mass is a third of that of all the guards is the subset exhibiting sadistic behavior. For one, delimiting the guards this way prerequisites knowledge of their individual volumes/masses. The original context of these sentences and their authors’ intent will likely remain a mystery by virtue of this being

- (43) un terz-o delle guardi-e aveva un comportament-o ritenut-o sadic-o
 a third-M.SG of.the.F.PL guard-F.PL had.3SG a behavior-M.SG consider-M.SG sadistic-M.SG
 ‘One-third of the guards exhibited behavior considered sadistic’
- (44) ne emerge che un quint-o delle donn-e inglesi vorrebbe un “design”
 EXPL comes.out C a fifth-M.SG of.the.F.PL woman-F.PL English.PL would.like.3SG a design
 nuov-o per [il propri-o naso]
 new-M.SG for the.M.SG own-M.SG nose
 ‘It turns out that one-fifth of British women would like a new “design” for their nose’¹⁹

To summarize, measure FPs in both novel and corpus data have been found to be compatible with predicates which range over (sums of) integrated wholes. I take this as strong evidence that measure FPs can pick out pluralities. They can do so for things besides pluralities of course (see “third of each”-cases discussed in §1), but this ability is left unexplained by Falco & Zamparelli’s suggested proposal. If the measure reading is resolvable to something like a universal grinder as they say, its application should bleed combination with a distributive predicate. This is because F&Z’s suggestion depends upon the destruction of the plural DP’s “individuation structure” in order to block the count reading. Distributive predicates and the like rely on this individuation structure to range over their arguments, thus predicting a ban on the cooccurrence of measure FPs and distributive predicates. As I have shown in (40)-(44), these predicted downstream effects are not realized, leaving us with a proposal which undergenerates.

F&Z themselves do not consider the possibility, but one may try to reconcile the above data with the topical proposal by calling upon another operator which sorts mass into individuated members; this would make the FP compatible with atom-sensitive predicates once again. It turns out such a mechanism has been proposed, more specifically the UNIVERSAL PACKAGER or UNIVERSAL SORTER (Bunt, 1981; Filip, 1999). Essentially serving as a covert classifier, it is most regularly invoked to capture cases where a canonical mass noun may exhibit properties of a count noun (e.g., bearing plural morphology and a cardinal modifier), as in (45). We may apply this operator to invert the effects of the universal grinder, as schematized in (46). In principle, this would permit the FP to combine with distributive and discrete collective predicates while simultaneously precluding the count reading.

(45) Jerry drank three beers last night

(46) $UP([terzo])(UG(\oplus[parete]))$

I consider this approach inadequate by virtue of Occam’s razor. Whereas F&Z’s original suggestion assumed only a single operator, explaining (40)-(44) involves positing another covert operator. This is unparsimonious as it is, but it also essentially duplicates one of our problems discussed above. If a packaging operator may apply to FPs to make them compatible with atom-referencing predicates, it does not seem to have any effect on the morphology of the FP; the FP (40), for example, has a singular denominator despite being the subject of distributive predicate *ellittico*. If this is the case, though, there is nothing stopping said packager from applying to any bare mass nominal. We end up predicting that bare mass nouns like *acqua* ‘water’ or *birra* ‘beer’ could be subjects of predicates which select for pluralities since there would be an accessible parse with with a silent packager ($UP([acqua/birra])$). This is not the case as far as I am aware, meaning we would have to arbitrarily stipulate a restriction of this operator to FP-arguments.

Allow me to conclude my commentary on Falco & Zamparelli’s proposal with a remark about these kinds of coercion processes and cross-linguistic variation. Count-mass coercion is indeed quite robust in English, the language whose data initially prompted the proposal of the universal grinder in Pelletier (1975). Its ubiquity puts the *universal* in *universal grinder*. Beyond the bounds of English, though, this alleged mechanism is not so universal. Perhaps most famously, Cheng et al. (2008) show that IW/atom-denoting nouns in Mandarin (e.g.,

corpus data, but the “100 out of 300” reading is reasonably the more likely of the two.

I maintain the position that these examples are genuine measure FPs. Recall in §1, fn. 1, I mention that measure and count readings can overlap depending if the members of the plural complement are congruent. Once we control for congruence, the two readings may be teased apart. What I conjecture is happening in (43)-(44) is the authors have assumed a context where individual variation along certain degree scales is negligible. Thus, the very same plurality which is a third of the guards by a non-cardinal measure like mass could also be a third of the guards by cardinality.

¹⁹Because the original was suggestive, I changed the last noun and the gender of the determiner, possessive adjective for concord reasons.

gǒu ‘dog’) do not permit mass-ish interpretations. Note that Mandarin is an articleless, numberless language, so we expect no morphological changes like those we see in English count-mass coercion.

- (47) *qiáng-shang dōu shì gǒu*
 wall-TOP all COP dog
 ✓ ‘There are dogs all over the wall’
 × ‘There is dog all over the wall’

[Mandarin]

Italian is a language with some nouns which, despite canonically being count, can also be coerced to a mass noun. This was seen for Italian *banana* in (36). However, fieldwork up to this point suggests that this is a particular property of certain nouns rather than a general property of Italian syntax-semantics. This is illustrated in (48), where two canonical count nouns like *scoiattolo* ‘squirrel’ and *calamaro* ‘squid’ do not permit mass readings.

- (48) #c’-è {scoiattol-o / calamar-o} sul paviment-o
 EXPL-be.3SG squirrel-M.SG / squid-M.SG on.the.M.SG floor-M.SG
 Int. ‘There’s squirrel/squid(-mass) on the floor’

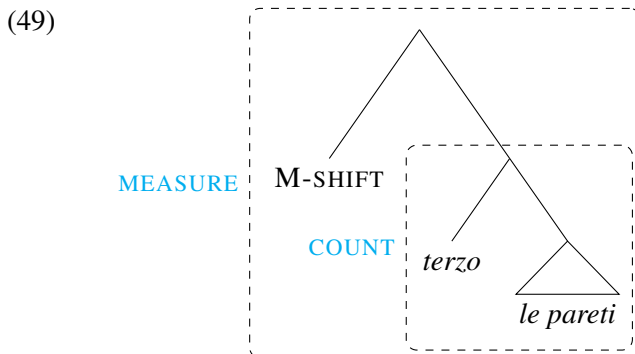
(EF)

While I stop short of a comprehensive investigation of Italian count nouns and their compatibility with measure FPs, measure FPs seem to be quite productive. We have seen them take pluralities countable, physical objects like *torte* and *pareti*, but also more abstract pluralities like *figure* and *film*. This casts an additional layer of doubt on F&Z’s proposal, since (i) the data in (48) might suggest a UG-type mechanism does not exist in Italian, and (ii) if the few mass-count coercions Italian does permit are indeed the work of such a mechanism, this should delimit the possibilities for measure FPs. This second point seems contrary to fact.

To summarize, the F&Z approach ends up under-generating for Italian. If measure FPs are actually just FPs which have taken a “massified” complement, they should be barred from combining with distributive or discrete collective predicates. As it turns out, such configurations of FPs and predicates appear to be possible in Italian. This is all made worse for the current set of data when we observe that count-mass coercion is not a very productive process for nominals in Italian. This examination has thusly yielded a new desideratum for an analysis: a measure FP must be able to refer to a plurality. Given the observations of this section, what seems to distinguish the count reading from the measure reading is how VMFs are defined rather than the mereotopological properties of the complement. I subsequently improve upon F&Z’ in the following section by changing the function of the operator.

6 Proposal II: Measure Reading via Measurement Parameter

My final proposal for an account of the topical alternation is very similar to F&Z’; I reduce the measure/count contrast to the presence or absence of a covert operator within the FP, respectively. Furthermore, I posit this operator bears features which are responsible for “denominator” agreement. However, I propose that the nature of this operator is quite different from the one F&Z have in mind. Rather than a grinder which modifies the mereotopological status of the complement, I reason that it is an intensional operator which takes the entire FP as its argument and permutes the parameter of evaluation by which variable measure functions are defined. For this reason it is called a MEASUREMENT SHIFTER (M-SHIFT). The two structures for FPs are schematized in (49).



Let me spell out transparently how this machinery works. Remember, Bale & Barner’s algorithm guarantees that pluralities will be counted rather than measured via their ordering of the set M by which VMFs are defined. I change this state of affairs by treating M as an intensional parameter by which logical forms are interpreted. As far as I am aware this is a novel development, and Bale & Barner themselves treat M as an ambient element of grammar/cognition. Doing so makes M shiftable, like other well-defined intensional parameters (worlds, times, contexts, and so forth). The guilty party in this case is M-SHIFT, which switches the “matrix” or default setting M to M' . M and M' have the same members, but the ordering relation is reversed—in this latter, cardinality is the maximal measure function.

I define M-SHIFT as in (50), taking a predicate of individuals and returning another where the set of evaluation is M' (51) rather than the “matrix” set M . As a result, measure functions in the scope of M-SHIFT will be defined with the Bale & Barner algorithm relative to M' instead of M . While M and M' are identical in terms of their members, the ordering relation has been reversed so that cardinality is now maximal with respect to other forms of measure. Per the algorithm, variable measure functions will be non-cardinal even if they are ranging over a plurality.

$$(50) \quad [[\text{M-SHIFT}]] = \lambda P_{\langle e,t \rangle} \lambda x_e. [[P(x)]]^{M'}$$

$$(51) \quad M' = \{\mu_{\text{weight}}, \mu_{\text{volume}}, \mu_{\text{mass}}, \mu_{\text{length}}, \mu_{\text{runtime}}, \dots, < \mu_{\text{card}}\}$$

Let us now walk through the derivation step by step. I assume a semantics for fractions partitives generally as shown in (52). This function takes a cardinal number n and returns a two-place predicate which is true if x is part of y and the degree of x is the degree of y divided by n .²⁰ I follow Wagiel in assuming that the preposition/genitive case is semantically vacuous. By default, the measurement parameter is set to M . The less-marked count reading proceeds as in (53)-(54), presumably with an additional step of existential quantification or application of a choice function to allow the structure to combine with predicates.

$$(52) \quad [[\text{FRAC}]]^{c,M} = \lambda n_n \lambda y_e \lambda x_e. x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{n}$$

$$(53) \quad \begin{array}{c} \text{(54-b)} \\ \swarrow \quad \searrow \\ \text{(54-a)} \quad \text{le pareti} \\ \swarrow \quad \searrow \\ 3 \quad \text{FRAC} \end{array}$$

$$(54) \quad \begin{array}{ll} \text{a.} & [[\text{terzo}]]^{c,M} = \lambda y \lambda x. x \sqsubset y \wedge \mu(x) = \mu(y) \times \frac{1}{3} \\ \text{b.} & [[\text{terzo delle pareti}]]^{c,M} = \lambda x. x \sqsubset \oplus [[\text{wall}]]^c \wedge |x| = |\oplus [[\text{wall}]]^c| \times \frac{1}{3} \end{array}$$

Once the fraction combines with a plurality, the ordering of M dictates that the VMF will be a cardinality function. Returning to the six walls of the studio apartment in §1, the conditions of (54-b) determine that the members of the extension will be the two-wall sums which are part of the wall plurality. The cardinality function is insensitive to individual variation in wall size, making any two-wall sum a possible referent of the count FP. This is schematized in *fig. 7*.

To get the measure reading, M-SHIFT applies to this structure and changes M to M' (55)-(56). By the same algorithm, VMFs in the scope of the operator will never be cardinality functions since it is now the maximal member of the set of MFs. Now, even a plurality can be measured along a non-cardinal dimension.

²⁰In Italian and many other languages, denominators (outside of suppletive forms like *half*) are derived from or exactly homophonous with the number’s corresponding ordinal. Given this generalization, it is fair to wonder whether the meaning of a fraction is derived from the meaning of an ordinal. Such an account for FPs has been given before (López Palma, 2011, 2012) but I abstract away from this detail since it has little bearing on the semantic alternation we are interested in.

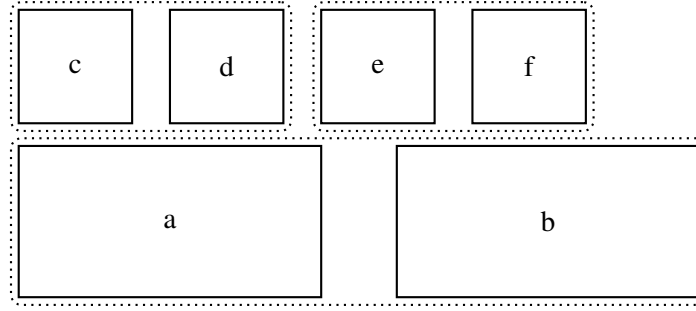
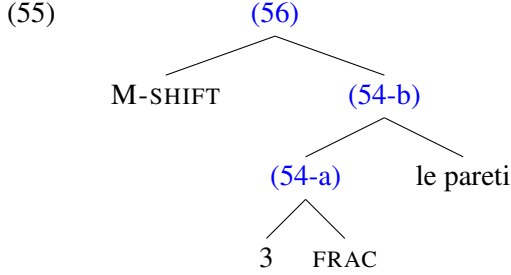


Figure 7: Partial extension of count FP (walls not to scale).



$$(56) \quad [[\text{M-SHIFT terzo delle pareti}]]^{c,M'} = \lambda x.x \sqsubset \oplus[[\text{wall}]]^c \wedge \mu(x) = \mu(\oplus[[\text{wall}]]^c) \times \frac{1}{3}$$

Let us imagine that in this case, context has determined that the relevant MF is a function of surface area. Given a plurality of walls whose cumulative surface area is $12h$ square meters, the resulting third must be $4h$ square meters. Ergo, the referent could be one of the large walls (a or b) or the sum of the small walls ($c \oplus d \oplus e \oplus f$), as in *fig. 8*.

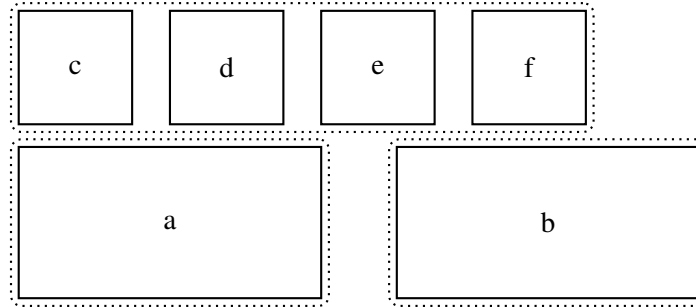
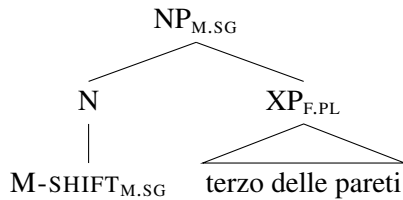


Figure 8: Partial extension of measure FP (walls not to scale).

As for how agreement works in this revised proposal, the story is about the same as what is sketched in §4. However, there are some details which I touch on here. As the grinder operator did in F&Z', M-SHIFT also enters the derivation in Narrow Syntax rather than post-syntactically at LF. This move is unproblematic despite the change in the operator's function since, essentially, this piece of machinery is an intensional operator and many proposed intensional operators transparently enter at NS (see doxastic predicates such as *believe*, von Fintel and Heim, 2011). By virtue of M-SHIFT being in the syntax means it may feed or bleed strictly-syntactic operations, like Agree. Additionally, the scope of the operator must change as compared to F&Z'. Where once the operator merged with the complement of the FP, now the operator must be above the fraction itself in order for VMFs to be evaluated by M' .

Similarly as before, I posit M-SHIFT comes with a masculine feature to derive the correct agreement pattern for measure FPs. Accordingly, a probe will copy said feature over the features of the complement by virtue of locality (57). However, I go further by suggesting that M-SHIFT is on some level a nominal. Doing so introduces a nominal syntactic category to the left-edge of the FP which, on either account of complement agreement I sketched above, the count FP structure lacks. This is motivated by the observation that certain nominal modifiers on the fraction preclude sentential agreement with the complement. In (58), for example, only masculine singular agreement is possible in the matrix clause when there is a restrictive relative clause modifying the fraction.

(57)



- (58) Il [terz-o del-le paret-i dipint-o da John] {è ross-o / *sono ross-e}
 the.M.SG [third-M.SG of-the wall-F.PL painted-M.SG by John] {be.3SG red-M.SG / be.3PL red-F.PL}
 ‘The third of the walls painted by John are red’ (EF)

To summarize the claims of this work cumulatively, I resolve the count-measure alternation to a covert piece of functional structure which (i) is nominal, bearing a gender feature, and (ii) shifts a parameter *M* to manipulate how VMFs are defined. It is the presence of this functional item that transforms a count FP into a measure FP. An astute reader may wonder, though, whether the cline in markedness I posit is justified beyond the fact that F&Z assumed as such. For starters, it is clearly not the case that one reading is more semantically-marked than the other. To illustrate, let us consider a transparent case. Markedness at the level of interpretation can be thought of as strength of meaning, which is itself a function of how few possible worlds it is true in. For example, *Newman is a mailman* is weaker than *Newman is a sinister mailman*. This is because the former is true in all worlds where Newman is a mailman, whereas the latter is true only in worlds where Newman is a mailman *and* sinister (which is of course a subset of the former set). Correspondence between semantic markedness (stronger meaning) and morphosyntactic markedness (more words/morphemes) is the *modus operandi* many assume, something that has come to be known as HORN’S DIVISION OF PRAGMATIC LABOR (Horn, 2001; de Swart and Farkas, 2010).²¹

As I illustrated in §1, though, the two readings for Italian FPs are disjoint—neither is a subset of the other. Granted, there can be substantial overlap between the measure and count reading of an FP in contexts where the individuals which compose the relevant plurality are congruent; in these cases, a subplurality can be a third by mass or by cardinality. However, the disunity of the two readings is made clear when we control for this. Regardless, the circumstances are such that no reading is stronger than the other. Additionally, FPs in Italian are superficially identical within the DP domain regardless of the reading being count or measure. On both counts, we have no support for the markedness cline I posit.

I maintain my position in spite of this because it confers analytical benefits for typology. To begin with, treating the measure FP as the sum of the count FP and an operator yields a straightforward explanation for some of the crosslinguistic variation observed in §1. As I discuss there, American English lacks the measure reading altogether. Relatedly, agreement is necessarily with the complement if an FP contains a plural DP. If the measure reading and its corresponding agreement pattern resolve to M-SHIFT (this proposal), American English can be defined as a language which simply lacks the operator. Thus, the difference between this language and Italian, etc. is a simple matter of lexical variation. This approach has the additional benefit of accounting for gaps in FP interpretation. There are languages where plural-containing FPs have a measure and count interpretation (Italian), and there are languages where such FPs only have a count interpretation. To my knowledge, however, there are no languages where a language has access to the measure interpretation to the exclusion of the count interpretation. Deriving the measure reading with an operator which is itself parasitic to the count reading makes these languages impossible, since the existence of the measure reading entails the existence of a count reading.

Furthermore, there are languages where the measure reading can be overtly more marked than the count reading. It is seen in (8) that Russian exhibits an agreement-interpretation similar to that in Italian; however, this language additionally exhibits a FP construction which includes the part-word *čast*’. Notice in (59) that only the measure interpretation is available and agreement is necessarily with the fraction/čast’. In light of these facts, we might interpret *čast*’ as an overt realization of M-SHIFT.²²

²¹...but this is not universally adopted. Perhaps most famously, plurals have been argued to be semantically-less marked than their singular (bare) counterparts in English and other languages (Sauerland et al., 2005).

²²This point requires some qualifications. Despite appearances (and how it is glossed), I do not take this use of *čast*’ to mean *part* in the more-familiar sense. Some languages have superficially similar FPs (i.e., ones which include an overt *part*-word) yet they exhibit completely different behavior. English has (or at one point had) a construction similar to Russian *čast*’ FPs, exemplified in (i). It is difficult to comprehensively evaluate the semantic import of *part* in these FPs because they seem quite marginal in contemporary Englishes; as of 2025, the Corpus of Contemporary American English (CoCA, Davies 2010) contains 7613 instances of *a third of* but only 23 instances of *a third part of*. Similarly, the British National Corpus (BNC, Davies 2004) contains 949 instances of *a third of* but a mere 3 instances of *a third part of*. Semantics aside, (i) shows that at minimum the presence of *part* does not preclude agreement with

- (59) tret'ja čast' sten krasn-{\check{a}ja/*ye}
 third.ADJ.F.SG part.F.SG wall.F.PL red-{F.SG/PL}
 'A third part of the walls are red'
 \check{M}, \times C

[Russian] (GL, VO)

7 Concluding Discussion

In conclusion, this work identifies an agreement-interpretation alternation found in several Indo-European languages and provides an account for Italian specifically. A novel contribution to the description of these items, it is reported that these readings can be disambiguated with certain agreement configurations; agreeing with the features of the FP's complement mandates a count reading, while copying the features of the fraction permits at least a measure reading. Ultimately, I explain the agreement facts as the result of an operator intervening between the probe and complement of the FP; the operator itself is defined as an intensional operator which changes how measure functions are defined at LF. This proposal is developed to also explain typological facts (why American English lacks the measure reading but no known language represents the reverse case). I consider an alternative from Falco & Zamparelli which attempts to derive the data through the application of a semantic type-shifter are considered, but I present a series of arguments showing that they do not provide a satisfying account.

Outside of the empirical contribution and its corresponding analysis, a major theoretical contribution of this work lies in measurement. Of the two readings, measure is deeply mysterious, given the discussion in §2. As I write there, a variable measure function which takes a plurality as its argument is traditionally defined as a cardinality function. This position has been a long-standing one of degree semantics; Bale & Barner et seq.'s finding that comparative constructions which range over pluralities robustly make evaluations by cardinality is a convincing reason. On the face of it, the mere fact that the measure reading exists seems to spell trouble for this tenet. In light of this, Falco & Zamparelli's suggestion for deriving the measure reading is particularly attractive. If they are right, we may maintain standard assumptions about how VMFs; to them, the complements of measure FPs are not genuine pluralities, so they cannot be counted. As I argue, though, there are reasons to think that measure FPs may indeed refer to pluralities.

I ultimately derive the facts by (i) treating the set of possible measure functions as a parameter which can be shifted, and (ii) introducing intensional operators in Narrow Syntax to shift them. This is admittedly a great deviation from established thought on degree semantics; however, I defend it as the best analytical option. In the face of FP data in Italian, etc., one could take the nihilistic position that there are no constraints on VMF-definition besides monotonicity—everything is purely contextual, regardless of the mereo-topological status of the argument. This is obviously wrong given the observed robustness of cardinality in comparative constructions, both across semantic type and across languages. This is also wrong in the microcosm of Italian FPs; the measure reading does not arise arbitrarily but is constrained by a particular configuration in positions of agreement. As I see it, my proposal is a compromise between the status quo that Bale & Barner establish (too strong, undergenerates) and what we might call METROLOGICAL NIHILISM (too weak, overgenerates). Measuring a plurality via a VMF is possible, but it is contingent on M-SHIFT—which is not present in all of the world's languages, if my theory of cross-linguistic variation is correct. Otherwise, the “matrix” ordering for measure functions is used.

As far as I know, there are no prior suggestions of this type in the literature, and skepticism is warranted. While the measurement-shifting story may improve upon, e.g., Falco & Zamparelli's analysis, it is fair to wonder if this is the best answer given its degree of specialization and, conversely, its lack of generalizability to other phenomena. For this analysis to hold water, it seems necessary to identify other phenomena which (i) are resolvable to VMFs, (ii) mandate other forms of measurement over counting, and (iii) are necessarily products of elements introduced in Narrow Syntax (as opposed to, e.g., the context). Whether or not these items exist is the object of future research.

a complement, unlike in Russian.

(i) [A] third part of the waters **were** turned into wormwood

(CoCA, [Davies 2010](#))

The details of English FP-*part* constructions are interesting and warrant further investigation. Chief among these questions is what the semantic difference between this and bare FPs might be, if any. However, it is beyond the scope of the present work. The agreement facts are enough to dissuade us from an analysis where the measure reading arises from the addition of *part*-words.

Acknowledgments

This research has benefited from conversations with many people, but I am especially appreciative of Liz Coppock, Kai von Fintel, Enrico Flor, Danny Fox, Martin Hackl, Nina Haslinger, Elise Newman, and Donca Steriade for their guidance. I would also like to thank audiences at the 2024 *Puzzles of Agreement* workshop, LSA 2026, and various MIT talk series for their insight and feedback. For their judgments and patience, I am indebted to Enrico Flor, Lorenzo Pinton, Genia Lukin, and Vlad Orlov. Errors are solely my responsibility.

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